

RESPONDING TO CYBERCRIME: PREVENTION STRATEGIES

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30.04.2026

Outline

1. Approaches to crime prevention: primary, secondary and tertiary crime prevention
2. Primary prevention
3. Secondary prevention
4. Tertiary prevention
5. Focus on ML and AI as crime prevention strategies

1. Approaches to crime prevention: primary, secondary and tertiary crime prevention

What is crime prevention?

- strategies and measures that seek to reduce the risk of crimes occurring, and their potential harmful effects on individuals and society, including fear of crime, by intervening to influence their multiple causes (United Nations, 2002)
- any activity by an individual or group (public or private), which attempts to **eliminate crime prior to it occurring or before any additional activity results**
- different ways of categorizing crime prevention, the main one adopts a public health approach focusing on preventing ill health by encouraging healthy lifestyles, rather than waiting for illness (investing funds early to save costs later)

Primary crime prevention (universal)

- is the most universal forms of crime prevention, being largely indiscriminating and targeting **society as a whole**
- primary prevention is intended **to target and prevent criminal behaviour before it occurs**
- primary prevention interventions involve **reducing crime opportunities** or **enhancing social factors that reduce an individual's likelihood of becoming involved in crime**
- herein analysed and assessed by focusing on Situational Crime Prevention and Universal Communication Strategies

Secondary crime prevention (at risk groups)

- Targets population groups who have not yet engaged in criminal behaviour but are at risk of doing so, i.e. **at risk groups** (e.g. kids showing signs of deviant/criminal behaviour)
- Focus on people having an increased proclivity for crime, but **before they graduate into more serious offending**
- herein analysed and assessed by focusing on Educational Workshops, Mentoring Programs and Targeted Warnings and Cautions

Tertiary crime prevention (known offenders)

- targets individuals after they have become involved in crime
- focus is on activities aimed at preventing those who have already engaged in criminal activity from future reoffending
- individuals are formally referred to programming by the criminal justice system following a criminal conviction
- herein analysed and assessed by focusing on Positive Diversions and Restorative Justice Practices

2. Primary cyber-crime prevention

Defining Situational Crime Prevention

- definition by Clarke: “comprises **opportunity-reducing measures** (1) directed at highly specific forms of crime, (2) involving the management/design/manipulation of the immediate environment in as specific and permanent way as possible (3) so as to increase the effort and risk of crime and reduce the rewards as perceived by a wide range of offenders”
- key assumption: many offenders are rational and weigh up the costs and benefits derived from offending
- builds on the rational choice offender approach, the routine activities theory and the crime pattern theory

The 25 situational crime prevention techniques

Increase the Effort	Increase the Risks	Reduce the Rewards	Reduce Provocations	Remove the Excuses
1. Target Harden - Steering column locks - Anti-robbery screens - Tamper-proof packaging	6. Extend guardianship - Take routine precautions 'Cocoon' neighbourhood watch	11. Conceal targets - Off-street parking - Gender-neutral phone directories - Unmarked bullion trucks	16. Reduced frustrations and stress - Efficient queues and polite service - Expanded seating - Soothing music/muted lights	21. Set rules - Rental agreements - Harassment codes - Hotel registrations
2. Control access to facilities - Entry phones - Electronic card access - Baggage screening	7. Assist natural surveillance - Improved street lighting - Defensible space design - Support whistleblowers	12. Remove targets - Removable car radio - Women's refuges - Pre-paid phone cards for pay phones	17. Avoid disputes - Separate enclosures for rival soccer fans - Reduce crowding in pubs - Fixed cab fares	22. Post instructions 'No parking' 'Private property' 'Extinguish camp fires'
3. Screen exits - Ticket needed for exit - Export documents - Electronic merchandise tags	8. Reduce anonymity - Taxi driver IDs 'How's my driving?' decals - School uniforms	13. Identify property - Property marking - Vehicle licensing and parts marking - Cattle branding	18. Reduce emotional arousal - Controls on violent pornography - Enforce good behaviour on soccer field - Prohibit racial slurs	23. Alert conscience - Roadside speed display boards - Signatures for customs declarations 'Shoplifting is stealing'
4. Deflect Offenders - Street closures - Separate bathrooms for women - Disperse pubs	9. Utilise place managers - CCTV for double-decker buses - Two clerks for convenience stores - Reward vigilance	14. Disrupt markets - Monitor pawn shops - Controls on classified ads - License street vendors	19. Neutralise peer pressure 'Idiots drink and drive' 'It's OK to say no' - Disperse troublemakers at school	24. Assist compliance - Easy library check-out - Public lavatories - Litter bins
5. Control tools /weapons 'Smart' guns - Disabling stolen mobile phones - Restrict spray paint to juveniles	10. Strengthen formal surveillance - Red light cameras - Burglar alarms - Security guards	15. Deny benefits - Ink merchandise tags - Graffiti cleaning - Speed humps	20. Discourage imitation - Rapid repair of vandalism - V-chips in TVs - Censor details of modus operandi	25. Control drugs and alcohol - Breathalysers in pubs - Server intervention - Alcohol-free events

Are SCP measures effective in preventing cybercrime?

Focus on five measures:

1. *Antivirus software*: effective strategy
2. *Warning messages*: see factsheet **What works in situational cybercrime prevention?**
3. *Primary Crime Prevention Apps*: many typologies of apps, no evaluation: see article **Primary Crime Prevention Apps: A Typology and Scoping Review**
4. *Internet Vigilantism*

Defining Universal Communication Strategies & cybercrime

- Mass media messages (broadcast on tv or through other audio/video channels) that aim to deter people from committing crimes
- Two key primary mechanisms: 1) changing perceptions of the risks of offending; 2) promoting positive behaviours and values
- + special category: appeal to third parties to intervene
- Very little has been done to tackle cybercrime so far, with few exceptions
- **Cyberchoices** <https://nationalcrimeagency.gov.uk/what-we-do/crime-threats/cyber-crime/cyberchoices>



3. Secondary cyber-crime prevention

Educational workshops

- bring together groups of (potential) offenders in single/multiple sessions, to educate them about the nature of crime, and reinforce prosocial values and behaviours
- deployed within school curricula, or may selectively target at-risk adolescents
- didactic + interactive elements
- rely on social-cognitive theories of learning
- mainly aimed at youth in primary or secondary schools to affect early onset offending behaviours

Educational workshops applied to cybercrime

To date, minimal empirical research on the application of workshops to cybercrime:

- i-SAFE program <https://isafeventures.com/>

2006 evaluation: “*There were **positive and significant changes in knowledge** between the treatment and comparison groups, both on average and over time. For the most part, there were **no significant changes in behavior** between the treatment and comparison groups*”.

Educational workshops applied to cybercrime

- Arizona Attorney General's Social Networking Safety Promotion and Cyberbullying Prevention (SNSPCP)
- Media Heroes

Mentoring

- involves the 'commitment of time and specific efforts by a more experienced person to the development of mutually beneficial, supportive and nurturing relationship with a less experienced person'
- corrective experience and intervention for at-risk youth who have experienced poor relations with parents or other caregivers
- theoretical underpinnings: importance of positive and supportive interpersonal relationships that offer guidance to children and adolescents throughout their development

Mentoring

- targets for mentoring interventions include young people who either exhibit a history of delinquent behaviour or characteristics that are assessed as at risk for juvenile delinquency (individual and environmental risk factors)
- so far, no evaluation research examining the utility of mentoring in relation to cybercrime
- lessons from studies conducted in relation to offline crimes

Targeted Warnings and Police Cautions

- used by police to warn potential offenders with the aim of deterring them from future offending
- stand-alone warnings, cautions, reprimands, and instructions to 'cease and desist', which may be carried out by law enforcement officers in person or through letters
- used also in relation to cybercrime e.g. UK investigation into Blackshades; LizardStresser booter service; international law enforcement operations
- all aimed at people already involved in cybercrime or at least on the periphery

4. Tertiary cyber-crime prevention

Positive diversions

- redirect individuals towards prosocial behaviours and peer influences; focus on rehabilitation
- usually for young people and 1st time offenders
- can take a variety of forms, including art, sports, and outdoors programs
- theoretical underpinnings:
 - imposing traditional criminal justice sanctions on young people has little effect on reoffending
 - positive diversions lean on social bond theories of crime

Positive diversions

- May be a substitute for criminal sanctions or court proceedings or go together with them
- Anecdotal evidence for cybercrime
- Widely deployed in response to various low-severity crimes
- Are they effective? Mixed findings
- A special category are those programs attempting to transform criminal behaviour into noncriminal behaviour (e.g. young person engaging in graffiti diverted in a program encouraging legal 'urban art') – relevant for cybercrime too

Positive diversions & cybercrime

- So far, no empirical evaluation of positive diversion programming for preventing cybercrime
- it is plausible that one could redirect the skills used for cybercrime into prosocial avenues (e.g. cybersecurity)
- graffiti artists and cybercriminals have something in common (sense of identity and pleasure from crime; status culture)

Restorative justice

- voluntary, community-based response to criminal behaviour that attempts to bring together the victim, the offender, and the community, in an effort to address the harm caused by the criminal behaviour
- strongly linked to Braithwaite's theory of re-integrative shaming
- can occur prior to, instead of, during, or after the formal criminal process
- tend to be single meetings of varying duration; chaired by a mediator; production of an agreement between the parties

Restorative justice & cybercrime

- there have been no studies to date on restorative justice's application to cybercrime
- wealth of evidence in the offline context; mixed findings
- application to cybercrime could help convey a message of the harmfulness arising from such crimes + some victims of cybercrime expressed interest
- case study: online hate speech (Finland)
<https://yle.fi/aihe/artikkeli/2019/11/06/face-to-face-with-a-man-who-wished-an-asylum-seeker-would-kill-my-family-the>